

Regime-Based SPY Strategy

Statistical Jump Model + Absorption Ratio

Main model: v5 Hybrid | Evaluation: Jul 2022 - Dec 2025

1. What We Did

We built a regime-switching trading strategy on SPY. The model reads daily market signals, labels each day as Bull (invested) or Bear (cash), and compares strategy performance to Buy & Hold. Several versions were tested; v5 is the chosen main model. v2 and v6 are alternative designs kept for comparison.

2. Data Used

- Primary file: Data/etf_ohlcv_20160301_20251231.csv - daily OHLCV, Mar 2016 to Dec 2025.
- 13 ETFs: SPY (traded asset), QQQ, RSP (breadth/momentum), plus 10 sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLRE, XLU, XLV, XLY) for Absorption Ratio.
- Returns use Adj_Close (split/dividend adjusted). Log returns are computed day-over-day.
- Out-of-sample (OOS) evaluation window: 1 Jul 2022 to 31 Dec 2025 (src/config.py). All versions are clipped to this window so metrics are comparable.
- S&P 500/400/600 PIT stock files exist in Data/ but were not used in the final pipeline - sector ETFs drive the Absorption Ratio instead.

3. Step-by-Step Procedure

- Step 1: Load ETF prices and compute log returns for all 13 tickers.
- Step 2: Compute Absorption Ratio (AR): rolling PCA on 10 sector return correlations (126-day window, 2 components). Higher AR = more correlated sectors = fragile market.
- Step 3: Compute AR z-score: fast (15-day) vs slow (126-day) rolling mean/std of AR.
- Step 4: Build feature matrix for the Jump Model (JM): SPY downside vol, Sortino ratios, vol ratio, breadth (RSP vs SPY), and optionally AR columns depending on version.
- Step 5: Walk-forward cross-validation: each month, refit JM on the prior 3 years (~756 days), pick the best lambda from the candidate grid, predict Bull/Bear for that month.
- Step 6: Backtest: Bull -> hold SPY (full or graded exposure); Bear -> cash (0% SPY). Signals are lagged 1 day - no look-ahead.
- Step 7: Report metrics: CAGR, volatility, Sharpe, max drawdown, Calmar, % time invested. Save equity curve plots and CSV outputs.

4. Jump Model (JM)

The Statistical Jump Model (Shu, Yu & Mulvey, 2024) splits the market into 2 states (Bull and Bear). Each

day it assigns the day to the state whose centroid best fits that day's feature vector - but switching states costs a penalty λ .

Objective: minimise sum of squared fit errors + $\lambda \times$ (number of regime switches).

Higher λ -> stickier regimes, fewer switches, slower to detect a new Bear.

Lower λ -> more switches, faster reaction, but can flicker and overfit.

Each month, walk-forward CV picks the λ that best separates Bull vs Bear days on the prior 3-year training window (using SPY return separation as the score).

5. Main Model - v5 Hybrid

v5 combines two ideas:

(A) v1-style JM regime - trained on 7 features including AR inside the feature matrix. This gives a stable, sticky Bull/Bear label (~7 regime switches over OOS).

(B) v4-style AR overlay - applied separately at backtest time. Even when JM still says Bull, high/rising AR z-score can cut exposure or force a fast exit (handles late Bear detection, e.g. 2025-style lag).

Position sizing (v5 hybrid):

- JM Bear -> 0% (cash)
- JM Bull + AR calm -> 100% SPY
- JM Bull + AR fragile -> 75% floor (bull_floor), can scale down further on exit rules
- AR exit: z-score > 1.25, or high + rising for 3 days -> drop to cash

v5 OOS results (Jul 2022 - Dec 2025):

JM baseline alone: CAGR 10.4% | Sharpe 0.96 | Max DD -12.5% | Invested 72.5%

v5 Hybrid: CAGR 6.7% | Sharpe 0.70 | Max DD -12.5% | Invested 66.0%

Buy & Hold: CAGR 18.5% | Sharpe 1.10 | Max DD -19.2% | Invested 99.8%

The hybrid trades some return for faster AR-driven exits and lower time in market.

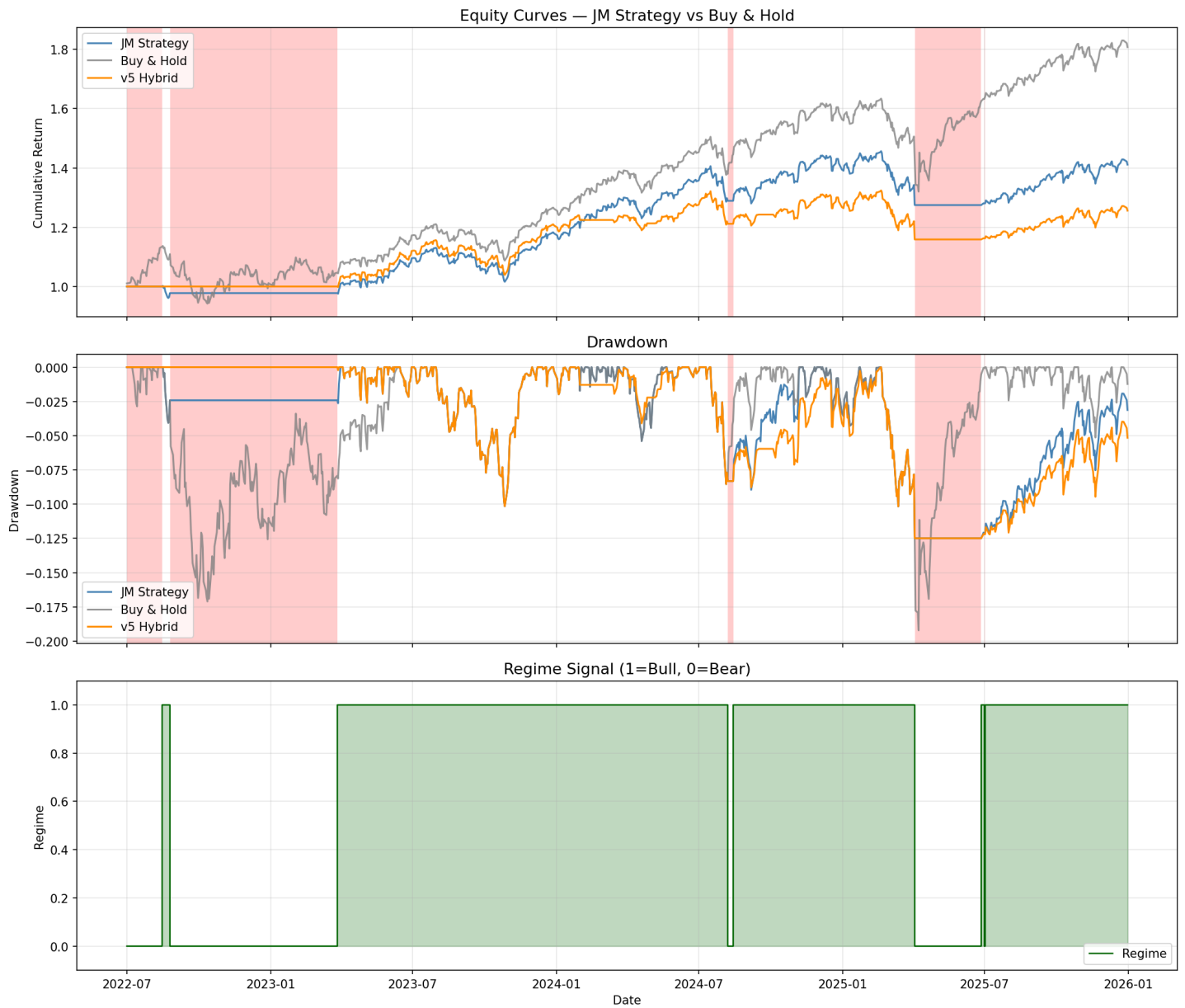


Figure 1 - v5: JM Strategy (blue), v5 Hybrid (orange), Buy & Hold (grey)

6. Alternative Models

v6 - AR-first Cascade

JM is trained on 5 SPY microstructure features only (no AR in JM). AR acts as a pre-filter at backtest time - checked before JM:

1. AR veto ($z > 1.0$, or high + rising for 3 days) -> cash (0%)
2. Else JM Bear -> cash
3. Else JM Bull and no AR veto -> 100% SPY

Strict binary 0/1 sizing - no partial positions.

v6 OOS: JM CAGR 8.4%, Sharpe 0.82, Max DD -11.8% | Cascade CAGR 4.9%, Sharpe 0.56, Max DD -10.4%, Invested 58.7%

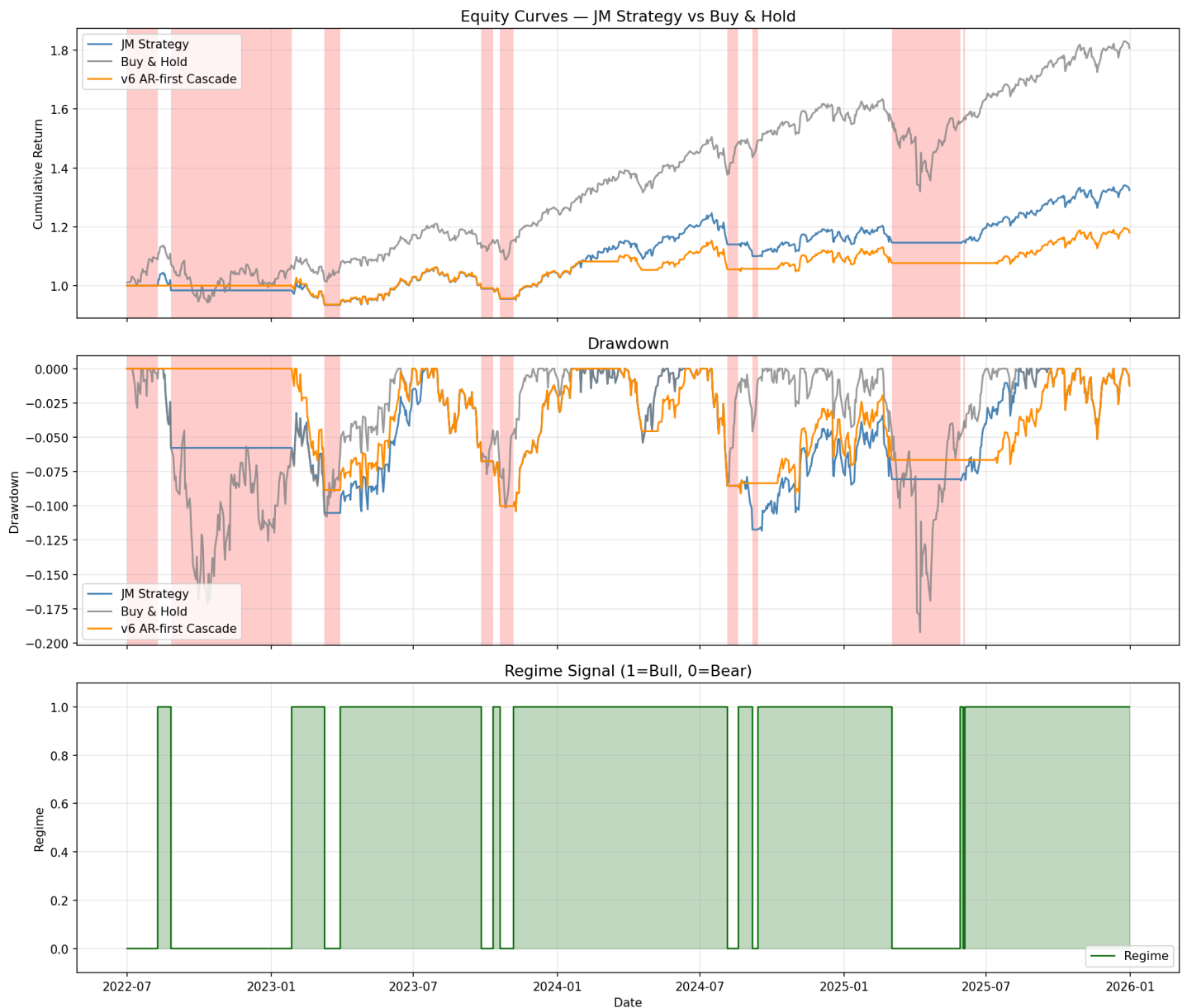


Figure 2 - v6: JM Strategy vs AR-first Cascade vs Buy & Hold

v2 - 2x2 Grid (JM trend x AR fragility)

JM and AR are kept on separate axes and combined in a 2x2 table:

	AR stable	AR fragile
JM Bull	100%	50%
JM Bear	0%	0%

JM uses 5 microstructure features (no AR). AR z-score > 1.0 marks 'fragile'. Bull + fragile days get half exposure instead of full.

v2 OOS: JM CAGR 8.4%, Sharpe 0.82 | 2x2 Grid CAGR 6.6%, Sharpe 0.72, Max DD -10.4%, Invested 71.8%

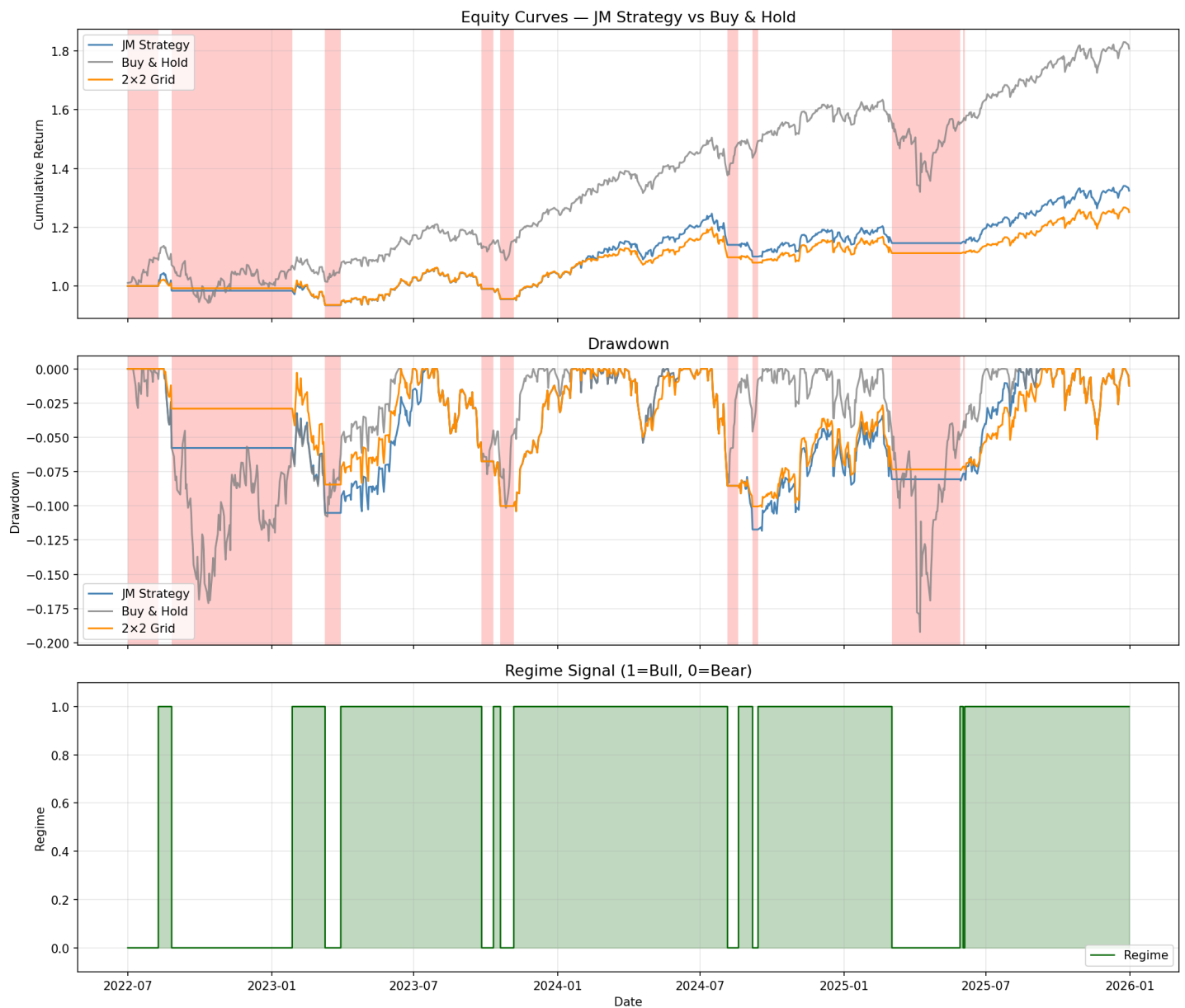


Figure 3 - v2: JM Strategy vs 2x2 Grid vs Buy & Hold

7. Lambda Tuning

Lambda is the jump penalty in the JM. It controls how sticky regimes are.

- Early runs used candidates starting at 5 and above only: [5, 10, 15, 20, 25, 30, 40, 50].
- Later, lambda = 3 was added to the grid: [3, 5, 10, 15, 20, 25, 30, 40, 50].
- Walk-forward CV picks one lambda per month. In practice, lambda = 3 was chosen most often once it was allowed - especially when the model needed faster regime switches.
- Meaning: pushing lambda as low as the grid allows makes the model switch regimes more easily. That fixes late Bear detection but can cause flickering if taken too far (values 1-2 were tested in lambda experiments and rejected as too noisy).

8. Issues Faced

Issue 1 - Sticky regimes / late Bear detection

When AR was included inside JM features (v1), regimes became very sticky - only ~7 switches over the full OOS window. The model stayed Bull too long during drawdowns (visible in the 2022 crash: market fell before the red Bear shading started). AR inside features reinforced stickiness because AR itself moves slowly.

Issue 2 - Lambda gravitating to the minimum

Once lambda = 3 was allowed, monthly CV often picked the lowest available value. That is expected: lower lambda = cheaper to switch = more reactive Bear calls. The trade-off is return vs responsiveness - very low lambda (1-2) over-switches and hurts out-of-sample stability. Lambda = 3 became the practical floor.

9. Summary

v5 is the main model: sticky JM regime (7 features) plus a separate AR overlay for fast exits and graded re-entry. It reduces max drawdown vs Buy & Hold (-12.5% vs -19.2%) but gives up absolute return. v6 (AR pre-filter cascade) and v2 (2x2 grid) are cleaner alternative architectures explored along the way. Earlier attempts (v1, v3, v4, lambda experiments) are archived in Scrapped_Tries/.